

Two coupled ODEs

integrated jointly from $t = 0$ to $t = 1$

Pretrained flow matching model

$$v_{\theta}(x, t)$$

Primal · samples

$$\dot{x}_t = v_{\theta} - J_g^{\top}(\lambda_t + c g)$$

λ_t

$g(x_t)$

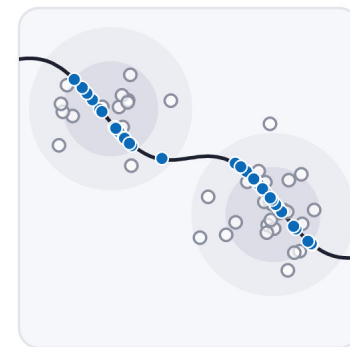
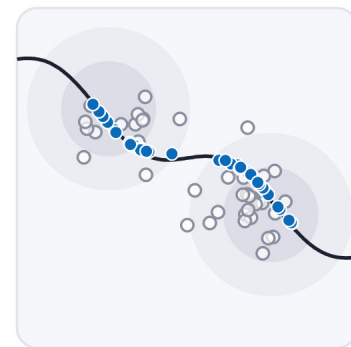
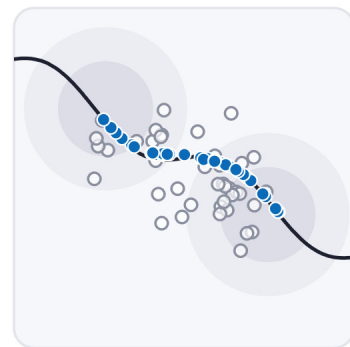
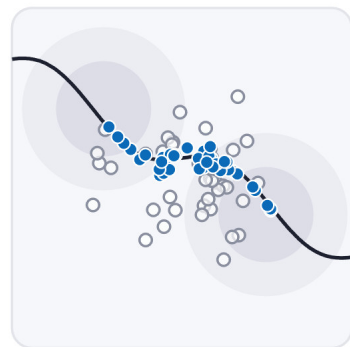
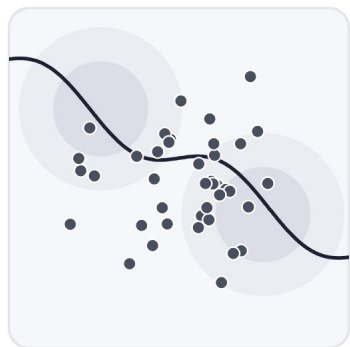
Dual · Lagrange multipliers

$$\dot{\lambda}_t = g(x_t) / (1 - t)^p$$

Constrained samples

$$g(x_1) = 0$$

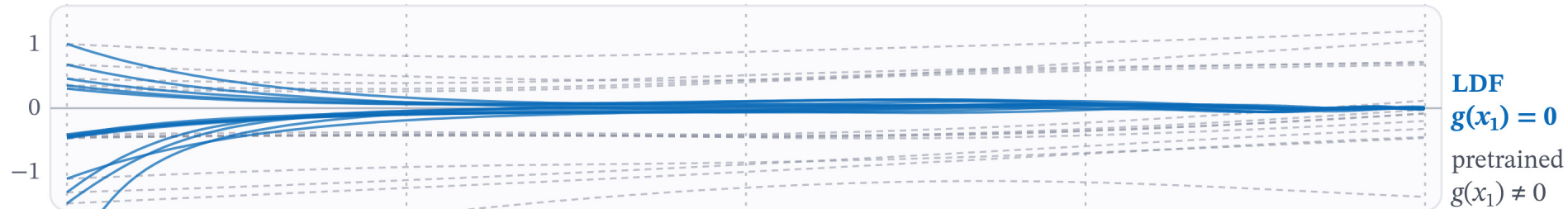
Primal samples x_t



— constraint $g(x) = 0$

○ pretrained flow ● LDF

Primal residual $g(x_t)$ feeds the dual



Dual Lagrange multiplier λ_t accumulates residual, pushes back on x_t

